

A Variational Quantum Circuit Framework for High-Fidelity Quantum State Preparation Using Quantum Neural Networks

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Abstract

Quantum state preparation (QSP) is a fundamental requirement in quantum computing, underpinning the accurate execution of quantum algorithms across applications such as quantum simulation, quantum machine learning, and quantum communication. This paper introduces a variational quantum circuit framework that integrates Quantum Neural Networks (QNNs) to address the problem of preparing target quantum states with high fidelity. The framework employs a structured variational ansatz paired with geometry-preserving loss functions, namely Trace distance, Fubini-Study distance, and Bures distance, that respect the intrinsic metric structure of quantum state space. To address practical challenges such as computational cost and vanishing gradients (barren plateaus), the framework uses a parameter shift rule for exact gradient estimation and a momentum-based adaptive optimizer tuned for quantum circuit landscapes. Experiments across 2- to 5-qubit systems demonstrate consistently high fidelities. Using Trace distance, fidelities of 0.999933, 0.999900, 0.997691, and 0.937918 were achieved for 2-, 3-, 4-, and 5-qubit systems, respectively, with training times ranging from 14.40s to 211.09s. With Fubini-Study distance, fidelities reached 0.999922, 0.999896, 0.999761, and 0.921357, with training times between 14.46s and 199.54s. With Bures distance, fidelities of 0.999940, 0.999930, 0.999261, and 0.942159 were obtained, with training times from 19.29s to 200.02s. Among the three, Bures distance as a loss function yielded the most consistent results across qubit counts and demonstrated greater robustness to noise and hardware imperfections compared to established baseline methods. These results show that the suggested paradigm for quantum state preparation is viable and scalable, relevant to quantum machine learning pipelines.

Keywords: Quantum State Preparation (QSP), Quantum Neural Network (QNN), Variational Quantum Algorithms (VQA), Quantum Native Loss Function, Gradient-Based Optimization in Quantum Circuits, Variational Ansatz Design.

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1. Introduction

Quantum computing (QC) has emerged as a transformative field with the potential to revolutionize various domains, including cryptography, optimization, machine learning(ML), and algorithm design [1] [2] [3]. A critical challenge in QC is the preparation of quantum states, which form the foundation for the execution of quantum algorithms [4]. QSP refers to the process of initializing a quantum system in a specific, desired state, and it plays a pivotal role in applications ranging from quantum simulation and QML to quantum communication and cryptography. The fidelity with which these quantum states can be prepared directly influences the performance and accuracy of quantum algorithms, making efficient and high-fidelity state preparation essential for the advancement of quantum technology [5].

Beyond QC, QSP is of paramount importance in quantum biology and quantum chemistry. In quantum biology, the processes by which biological systems operate on the molecular and atomic scales, such as photosynthesis, enzyme catalysis, and protein folding, may be based on quantum mechanical effects [6] [7] [8]. Understanding and manipulating these quantum states can lead to breakthroughs in the development of bio-inspired quantum technologies, thereby opening new frontiers in biotechnology and medicine. Similarly, in quantum chemistry, preparation of specific quantum states is crucial for simulating molecular systems and chemical reactions. QSP enables accurate modeling of complex chemical processes, which is vital in fields such as drug discovery, material science, and energy storage [9] [10].

As we continue to explore the broader applications of QSP, its role extends to QML and quantum communication. In QML, the precise preparation of quantum states allows efficient encoding of data in quantum circuits, facilitating tasks such as classification [11] and regression [12]. This enables quantum algorithms to leverage quantum parallelism, thereby leading to faster computation and improved model accuracy. Similarly, in quantum communication, accurate state preparation is essential for protocols such as quantum key distribution (QKD), which relies on entangled states to ensure secure communication. The ability to prepare these states with high fidelity is fundamental to the success of quantum communication systems, including encryption, quantum teleportation, and the development of quantum repeaters, all of which are key to building secure and scalable quantum networks [13] [14] [15].

Traditional QSP methods, such as gradient ascent pulse engineering(GRAPE) and chopped random-basis optimization (CRAB), have achieved notable successes in the preparation of simple quantum states [16] [14]. However, these techniques are often limited by issues such as susceptibility to local optima, scalability to large quantum systems, and sensitivity to noise in practical settings. Moreover, conventional approaches require complex pulse sequences and high computational overhead, which become increasingly impractical as the number of qubits in a quantum system increases.

Recent advances in ML and Deep learning (DL) have introduced a new paradigm for solving complex optimization problems in QC. Specifically, QNNs have gained significant attention as powerful tools for QSP. QNNs leverage the expressive power of DL techniques to optimize quantum circuits in a manner that exceeds the limitations of traditional methods [17] [18]. By learning the optimal QSP protocols through training on data, QNNs are capable of approximating complex quantum states with high fidelity while also being more robust to noise and hardware imperfections.

The application of QNNs to QSP offers several distinct advantages. First, QNNs are highly

adaptable to the intricate nature of quantum systems, where the *Hilbert space* grows exponentially with the number of qubits. This adaptability allows QNNs to navigate a vast, high-dimensional parameter space more efficiently than traditional optimization methods. Second, QNNs are capable of handling noisy intermediate-scale quantum (NISQ) devices, where errors and decoherence limit the performance of the quantum circuits. By incorporating noise resilience into the training process, QNNs can achieve high-fidelity state preparation even in the presence of imperfections in quantum hardware [19]. Despite the promising potential of QNNs, several challenges remain in their practical application to QSP [20]. The design of neural network architectures that can effectively represent complex quantum states is nontrivial, and the optimization of such models requires novel techniques to balance expressivity and trainability.

This research introduces a novel VQA architecture that incorporates a QNN to develop a high-fidelity QSP on NISQ devices. To enhance expressivity and scalability, our approach employs a novel ansatz that draws inspiration from neural networks and employs linear parameter scaling. The main contributions of our work are as follows:

- We develop a variational QNN-based quantum state preparation framework for NISQ devices that prepares arbitrary target states with high fidelity, maintains a trainable circuit architecture, and scales linearly in parameter count with qubit number.
- We incorporate three geometry-aware loss functions, namely Trace distance, Bures distance, and Fubini-Study distance, into a unified optimization framework that follows the natural metric structure of quantum state space and improves convergence under quantum constraints.
- We implement an enhanced parameter shift method for exact gradient computation, enabling reliable optimization of deep variational circuits while keeping computational overhead manageable.
- We evaluate the framework on state preparation tasks spanning 2 to 5 qubits, measuring fidelity, convergence behavior, and training time across system sizes to identify the best-performing loss function for each configuration.

This section elucidates the concept of QSP, its significance, and the obstacles associated with attaining high-fidelity state preparation for quantum computing applications. The subsequent section evaluates the current related work of QSP. Section 3 outlines the proposed VQA framework, detailing the integration of QNN, quantum native loss functions, and enhanced gradient computation techniques. Section 4 presents experimental results and validates the effectiveness of the proposed framework across various quantum systems. Section 5 concludes the paper with a summary of the contributions and suggestions for future research directions.

2. Related Work

This section reviews the most pertinent extant works on quantum state preparation and emphasises their key methodological contributions.

Chao-Chao et al. [21] presented a stochastic prediction (SP) approach to better prepare the quantum states in silicon quantum dots. The SP algorithm uses random predictions to improve

fidelity and overcome local optima, unlike conventional approaches such as gradient ascent pulse engineering (GRAPE), chopped random-basis optimization (CRAB), and the greedy algorithm (GA). This neural network has several important parameters, such as the total evolution time (4π for a single qubit and 10π for two qubits), action duration ($\pi/5$ for a single qubit and $\pi/2$ for two qubits), maximum time step (20 for both types), number of allowed actions (eight for a 1-qubit and sixteen for 2-qubits), and a learning rate of 0.0005 for a single qubit and 0.001 for 2-qubits. With a greater fidelity of 0.97 for a single qubit in noiseless situations, the SP algorithm routinely outperforms competing approaches. The resilience of the algorithm is demonstrated by the fact that fidelity drops in noisy settings but remains consistent (approximately 0.93) when trained with noise.

Ahmed et al. [22] presented a conditional generative adversarial network (QST-CGAN) designed for the reconstruction of optical quantum states. The network is composed of a generator that reconstructs physically valid density matrices using custom layers, and a discriminator that assesses the quality of these reconstructions. Both networks employ LeakyReLU activations, utilize Adam optimization with a learning rate of 2×10^{-4} , and implement binary cross-entropy loss. The QST-CGAN demonstrates markedly improved fidelity compared to conventional maximum likelihood estimation techniques, surpassing 0.99 in noiseless scenarios and consistently maintaining values above 0.95 in the presence of moderate noise, with fewer iterations and measurements required.

Melnikov et al. [23] presented an effective method for preparing quantum states through Riemannian gradient algorithms to optimize tensor networks and particular quantum circuits designed for hardware compatibility. They significantly simplify quantum encoding by utilizing Matrix Product States (MPS) to represent input vectors through analytical function sampling. The simulation findings indicated that the system possesses an exceptionally high preparation fidelity of ($>99.9\%$) and also exhibits polynomial runtime scaling.

Li et al. [24] empirically demonstrated quantum state reconstruction utilising generative models founded on recurrent neural networks (RNNs). This method is utilized on a superconducting transmon qubit platform to prepare and measure the Greenberger-Horne-Zeilinger (GHZ) and Bell states involving up to five qubits, employing informationally complete Pauli-4 POVMs. The authors presented the fidelities of the reconstructed quantum states, attaining high classical fidelities exceeding 99% and quantum fidelities of approximately 95%, which are comparable to or slightly lower than those of traditional QST methods. The fidelities obtained using traditional QST were 98.0% for 2-qubits, 97.9% for 3-qubits, 93.3% for 4-qubits, and 89.4% for 5-qubits.

Carrasquilla et al. [25] proposed a scalable and efficient method for quantum state reconstruction that integrates generative ML models with tensor network structures and neural network-based probabilistic models. Their approach efficiently recreates perturbed ground states and GHZ states of complicated many-body systems, including the 1D transverse-field Ising model and the 2D triangular lattice Heisenberg model.

Sierra-Sosa et al. [26] used TensorFlow Quantum to investigate the effects of QSP on QML. They tested three different models, namely, QCNN, angle hybrid, and amplitude hybrid, and found that the latter produced better results. By the end of the eighth epoch, the amplitude hybrid model outperformed the angle hybrid model by as much as 8.9% in terms of accuracy, with the former exceeding 90% accuracy at centroid distances, whereas the latter fell short of this mark.

Nicolas et al. [27] presented a method that uses neural networks to prepare arbitrary quantum states in qubit-coupled quantum harmonic oscillators (QHO). The neural network was used to forecast

the pulse parameters required to reach a target state, and the oscillator and qubit were both subjected to a series of square pulses as part of the suggested procedure. As the number of pulses in the sequence increases, the model’s performance improves, demonstrating excellent accuracy in state preparation with 99.9% fidelity for qubit states and 97% for qutrit states. More precisely, qubits exhibited optimal outcomes when pulsed with seven pulses, whereas qutrits exhibited optimal outcomes when pulsed with eight pulses. The created states exhibited high purity according to the model, as demonstrated in, with qubit states achieving fidelities of up to 99.99%.

Krisnanda et al. [28] introduced a new method for creating and concentrating quantum resource states using QNN. This method employs linear optical mixing, incoherent pumping, coherent pumping, and fermionic nodes in a quantum reservoir network. To produce high-fidelity quantum states, including cluster, discorded, NOON, and maximally entangled states, the network was trained using a hybrid random-genetic approach that optimized the mixing coefficients. For the maximally entangled NOON, W, and cluster states, the experiments show very high success rates of more than 99%, even when there is noise such as dephasing and depolarization.

Wang et al. [29] presented a neural network-based technique for QSP in noisy situations. The approach was assessed utilizing 1-qubit and 2-qubit systems, incorporating neural network-augmented control pulses to alleviate dynamic and quasi-static noise. The experimental results indicate that the method surpasses traditional techniques, including ADAM optimization, attaining fidelities of 99.5% for single-qubit and 98.6% for two-qubit state preparations. The model exhibited 94% fidelity under dynamic noise and 97% fidelity under quasistatic noise, accompanied by reduced preparation times and accelerated convergence.

Torlai et al. [30] investigated the use of restricted Boltzmann machines (RBMs) to recover quantum states from experimental data generated by a Rydberg quantum simulator, which is associated with noise. Based on the experimental results, the RBM successfully restores quantum states inside an 8-atom system with a 99% fidelities and more than 95% in a 9-atom system.

Arrazola et al. [31] demonstrated ML-based automation for quantum circuit design. Optical gates (including interferometers, displacements, squeezers, and Kerr gates) make up the Strawberry Fields simulator, which uses a continuous variable QNN optimized by gradient descent using TensorFlow. For state preparation, single-photon states had 99.998% fidelities, ON states 99.93%, GKP states 99.60%, random states 99.82%, and NOON states 99.90%.

Hanrui et al. [32] showed RobustState, a novel approach to enhance the accuracy of quantum state preparation on quantum devices subjected to noise. The RobustState was evaluated on a variety of quantum state preparation tasks, including arbitrary quantum states, partial differential equation (PDE) states, QML states, and quantum error correction (QEC) codewords. Based on the experimental results, there is a significant improvement in fidelity (a 50% increase for 4-qubit states and a 72% increase for 5-qubit states) and a reduction in coherence errors (up to 7.1 times more than current approaches).

He et al. [33] introduced a Revised Greedy (RG) algorithm for preparing universal quantum states, which attempted to address the drawbacks of the standard greedy (SG) approach by fixing its tendency to become stuck at local maxima. This method, usable in 1-qubit and 2-qubit systems, including silicon quantum dots and superconducting circuits, allows the production of arbitrary quantum states via dynamic control pulses. The results demonstrate that RG succeeds in achieving fidelities as high as 0.999 for single-qubit states in superconducting circuits and 0.977 in

semiconductor quantum dots. In both systems, the preparation of 2-qubit states led to fidelities exceeding 0.9.

Haimeng et al. [34] analyzed the sample complexity of neural network approaches for mixed-state reconstruction using Neural Quantum States(NQS). This study compared two NQS encodings: the Neural Density Operator(NDO) and Positive Operator-Valued Measurement NQS(POVM-NQS). As the state becomes more mixed, NDO loses some of its edges, but still works better for almost pure states. Conversely, POVM-NQS demonstrates uniform scaling, similar to standard shadow techniques, irrespective of the mixedness of the state.

3. Methodology

The formation of arbitrary quantum states in regulated quantum systems is a significant issue in quantum information science. This preparation facilitates applications in quantum simulations, QML, and algorithm initialisation. By utilising hybrid quantum-classical optimisation, VQAs have become a viable method for approximating goal quantum states using parameterized quantum circuits. This section offers a complete theoretical and mathematical study of the variational framework, covering the ansatz creation, formulation of the quantum native cost function, gradient evaluation, and optimisation approach. The workflow methodology of the proposed approach is illustrated in Figure 1.

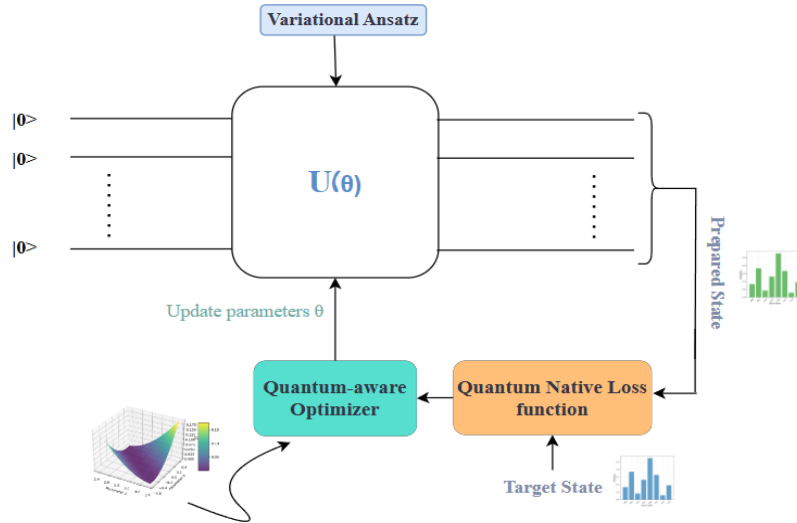


Figure 1: Proposed QNN-based variational framework for quantum state preparation.

3.1. Quantum System and Parameterized State Preparation

The Hilbert space of a quantum system with n qubits, where each qubit is linked to a two-dimensional complex vector space $\mathbb{C}^2$, is represented by $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$. The dimension of this composite Hilbert space grows exponentially with the number of qubits. Specifically,

$$\dim(\mathcal{H}) = d = 2^n \quad (1)$$

The computational basis $\{|i\rangle\}_{i=0}^{d-1}$, where each basis vector corresponds to an n -bit binary string $i = i_1 i_2 \dots i_n$ with $i_j \in \{0, 1\}$, forms an orthonormal basis spanning $\mathcal{H}$. These basis states satisfy the following conditions:

$$\langle i | j \rangle = \delta_{ij}, \quad \forall i, j \in \{0, \dots, d-1\} \quad (2)$$

where δ_{ij} is the Kronecker delta [4].

An arbitrary pure quantum state $|\psi\rangle \in \mathcal{H}$ can be expressed as a complex linear combination of these basis vectors:

$$|\psi\rangle = \sum_{i=0}^{d-1} \alpha_i |i\rangle \quad (3)$$

where the complex amplitudes $\alpha_i \in \mathbb{C}$ satisfy the normalization condition as follows:

$$\sum_{i=0}^{d-1} |\alpha_i|^2 = 1. \quad (4)$$

The global phase of $|\psi\rangle$ is physically irrelevant, and the state is fully characterized by the set $\{\alpha_i\}$.

Our primary goal is to design a parameterized family of unitary transformations $U(\boldsymbol{\theta}) : \mathcal{H} \rightarrow \mathcal{H}$, where $\boldsymbol{\theta} \in \mathbb{R}^M$ represents the vector of the real values of the tunable parameters, such that

$$|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})|0\rangle^{\otimes n} \quad (5)$$

approximates a predetermined target quantum state $|\psi_{\text{target}}\rangle$ with high fidelity.

The unitary $U(\boldsymbol{\theta})$ corresponds to a variational quantum circuit, or ansatz, composed of a sequence of parameterized quantum gates. The dimension M of the parameter space depends on the ansatz structure and depth, which determines the expressivity of the circuit in the approximation of complex states.

This framework is essential for VQA, wherein $\boldsymbol{\theta}$ is optimised through classical methods to minimise an appropriate cost function that measures the distance to $|\psi_{\text{target}}\rangle$.

3.2. Variational Ansatz Design

We consider a parameterized quantum circuit $U(\boldsymbol{\theta})$ acting on the *Hilbert space* $\mathcal{H} = (\mathbb{C}^2)^{\otimes n}$ of an n -qubit system, where the dimension of $\mathcal{H}$ is $d = 2^n$. The goal is to prepare a variational quantum state $|\psi(\boldsymbol{\theta})\rangle = U(\boldsymbol{\theta})|0\rangle^{\otimes n}$ that approximates a target quantum state $|\psi_{\text{target}}\rangle$ with high fidelity. Here, $\boldsymbol{\theta} \in \mathbb{R}^M$ denotes the vector of variational parameters to be optimized.

The ansatz structure was designed to balance the expressivity and trainability by alternating layers of single-qubit rotations and entangling operations. Specifically, the circuit consists of three layers of single-qubit rotations, each parameterized by rotations about the three Pauli axes X, Y , and Z . These are interleaved with two entangling layers composed of controlled-NOT (CNOT) gates arranged to maximize entanglement and connectivity among the qubits. Figures 2, 3, 4, and 5 illustrate the diverse ansatz structures with varying qubit counts employed in this study.

Mathematically, the unitary transformation implemented by the ansatz can be expressed as

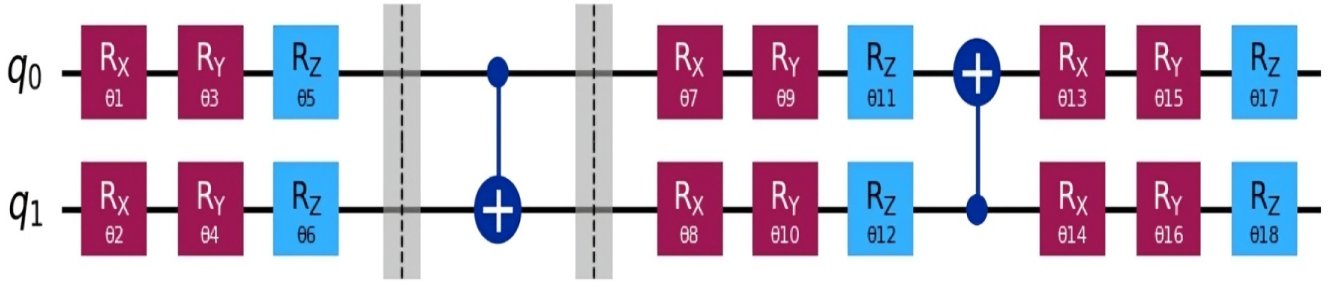


Figure 2: Random 2-qubit ansatz architecture.

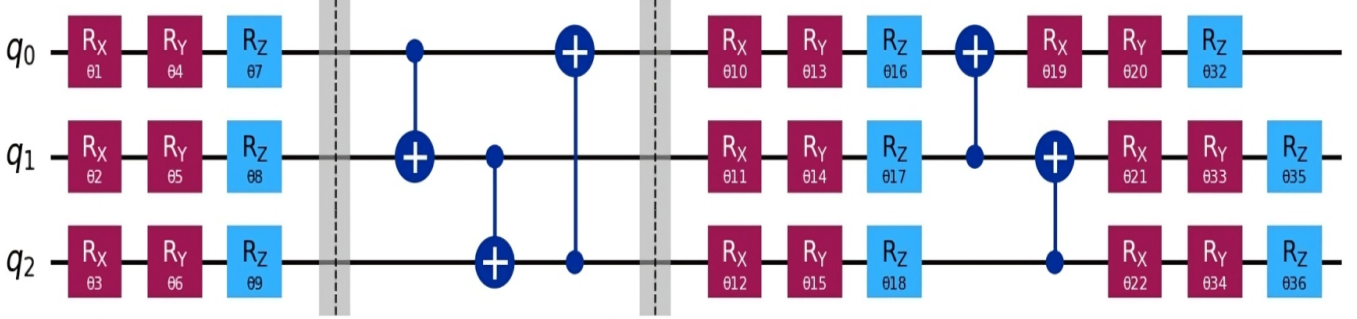


Figure 3: Random 3-qubit ansatz architecture.

$$\begin{aligned}
 U(\boldsymbol{\theta}) = & \left(\bigotimes_{i=0}^{n-1} R_X(\theta_{3i}) R_Y(\theta_{3i+1}) R_Z(\theta_{3i+2}) \right) E_1 \\
 & \left(\bigotimes_{i=0}^{n-1} R_X(\theta_{3n+3i}) R_Y(\theta_{3n+3i+1}) R_Z(\theta_{3n+3i+2}) \right) E_2 \\
 & \left(\bigotimes_{i=0}^{n-1} R_X(\theta_{6n+3i}) R_Y(\theta_{6n+3i+1}) R_Z(\theta_{6n+3i+2}) \right)
 \end{aligned} \tag{6}$$

where each single-qubit rotation is defined as:

$$R_\mu(\phi) = \exp \left(-i \frac{\phi}{2} \sigma_\mu \right), \quad \mu \in \{X, Y, Z\}$$

where σ_μ denotes the Pauli operator. The full parameter vector $\boldsymbol{\theta} \in \mathbb{R}^{9n}$ comprises the rotation angles applied to each qubit in each layer.

The entangling layers E_1 and E_2 are composed of CNOT gates arranged to improve qubit connectivity and entanglement generation. The first entangling layer E_1 consists of a linear chain of nearest-neighbour CNOT gates along with an additional CNOT connecting the last and first qubits, forming a circular entanglement pattern as follows:

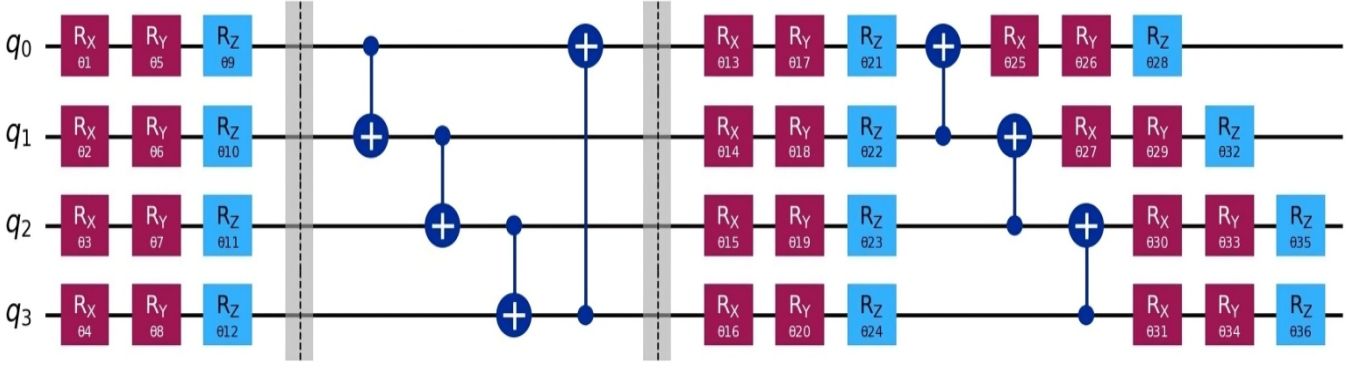


Figure 4: Random 4-qubit ansatz architecture.

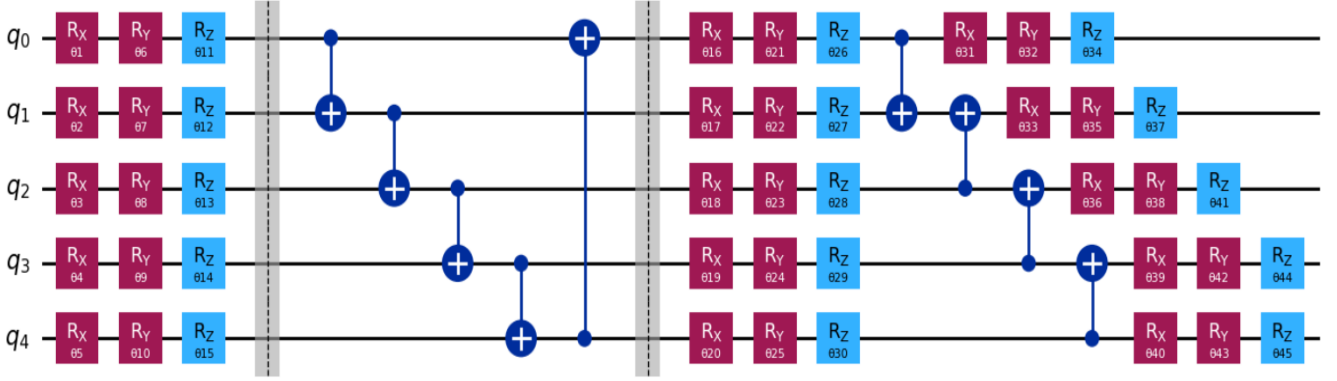


Figure 5: Random 5-qubit ansatz architecture.

$$E_1 = \prod_{j=0}^{n-2} \text{CNOT}_{j \rightarrow j+1} \cdot \text{CNOT}_{n-1 \rightarrow 0} \quad (7)$$

The second entangling layer E_2 reverses the direction of the CNOT gates to

$$E_2 = \prod_{j=0}^{n-2} \text{CNOT}_{j+1 \rightarrow j} \quad (8)$$

further enriches the circuit entanglement structure.

This architectural decision is driven by the necessity to produce intricate, multipartite entanglement, which is crucial for the precise representation of arbitrary quantum states. The rotations along all three Pauli axes ensure that any single-qubit unitary can be approximated over consecutive layers using the Euler decomposition theorem. The layered and bidirectional CNOT arrangement promotes a complex entanglement structure, enabling the ansatz to investigate an extensive subspace of the whole unitary group on n qubits. Significantly, the parameter count scales linearly with the number of qubits, $M = 9n$, providing a useful trade-off between resource requirements and expressive

capacity, making it appropriate for use in near-term quantum devices.

This approach empirically exhibits great fidelity in preparing complicated random quantum states, validating its appropriateness and adaptability for variational quantum state engineering applications.

3.3. Quantum Native Loss Functions

The optimization of variational quantum circuits for state preparation requires the meticulous selection of loss functions that accurately reflect the quantum geometric structure of the underlying *Hilbert space*. Conventional classical optimization methods frequently neglect the distinctive characteristics of quantum states, such as their normalization requirements, phase invariance, and the non-Euclidean geometry of the quantum state space. This study constructed and analyzed four unique quantum-native loss functions, each tailored to leverage various geometric characteristics of quantum state spaces to improve convergence and optimization efficacy.

3.3.1. Trace Distance-Based Loss Function:

The *Trace distance* provides an alternative quantum-geometric measure that captures the distinguishability between quantum states [4]. For pure states, the *Trace distance* can be expressed in terms of fidelity as follows:

$$D_{\text{tr}}(\boldsymbol{\theta}) = \sqrt{1 - F(\boldsymbol{\theta})} = \sqrt{1 - |\langle \psi_{\text{target}} | \psi_{\text{prepared}}(\boldsymbol{\theta}) \rangle|^2} \quad (9)$$

The *Trace distance*-based loss function is defined as:

$$L_{\text{trace}}(\boldsymbol{\theta}) = D_{\text{tr}}(\boldsymbol{\theta}) = \sqrt{1 - F(\boldsymbol{\theta})} \quad (10)$$

This formulation offers different optimization characteristics compared to the fidelity-based approach. The square root transformation magnifies minor variations close to the best solution, offering increased sensitivity in areas where fidelity nears unity. The enhanced gradient characteristics stem from the derivative framework, which contrasts with the fidelity loss, potentially circumventing specific local minima that hinder conventional fidelity optimisation. This loss function preserves metric consistency by directly aligning with the *Trace distance* metric in the realm of quantum states, thus assuring that the optimisation process adheres to the inherent geometric structure of the quantum state manifold.

3.3.2. Bures Distance-Based Loss Function:

The *Bures distance* represents a Riemannian metric in the space of quantum states, providing a geometrically natural measure for quantum state comparison. For pure quantum states, the *Bures distance* is given by:

$$d_{\text{Bures}}(\boldsymbol{\theta}) = \sqrt{2 \left(1 - \sqrt{F(\boldsymbol{\theta})} \right)} = \sqrt{2 \left(1 - \sqrt{|\langle \psi_{\text{target}} | \psi_{\text{prepared}}(\boldsymbol{\theta}) \rangle|^2} \right)} \quad (11)$$

The corresponding loss function is formulated as follows:

$$L_{\text{Bures}}(\boldsymbol{\theta}) = 2 \left(1 - \sqrt{F(\boldsymbol{\theta})} \right) = 2 \left(1 - \sqrt{|\langle \psi_{\text{target}} | \psi_{\text{prepared}}(\boldsymbol{\theta}) \rangle|^2} \right) \quad (12)$$

The distance-based loss function of *Bures* exhibits several distinctive properties that make it particularly suitable for quantum optimization. The optimisation path will follow geodesics in the correct metric space since the Riemannian structure takes into account the inherent geometry of quantum state manifolds. The loss function's invariance with respect to scale ensures uniform sensitivity over the optimisation domain, mitigating problems like undersensitivity in less optimal parts and oversensitivity in more optimal ones. The optimisation process is also linked to information-theoretic principles through this distance measure's relationship to the quantum *Fisher information metric*, which controls the fundamental limits in quantum parameter estimation theory.

3.3.3. *Fubini Study Distance Based Loss Function:*

The *Fubini Study metric* represents the natural Riemannian metric on complex projective spaces, making it particularly suitable for quantum state optimization where global phases are irrelevant. The *Fubini-Study distance* between quantum states is defined as:

$$d_{\text{FS}}(\boldsymbol{\theta}) = \arccos(|\langle \psi_{\text{target}} | \psi_{\text{prepared}}(\boldsymbol{\theta}) \rangle|) \quad (13)$$

The corresponding loss function is the following:

$$L_{\text{FS}}(\boldsymbol{\theta}) = \arccos^2(|\langle \psi_{\text{target}} | \psi_{\text{prepared}}(\boldsymbol{\theta}) \rangle|) \quad (14)$$

This loss function incorporates several mathematically elegant properties that distinguish it from other native quantum approaches. Projective geometry naturally accounts for the projective structure of quantum state spaces, recognizing that quantum states are elements of the complex projective *Hilbert space* rather than the *Hilbert space* itself. Phase invariance explicitly ignores the global phase differences between states, focusing on the optimization of physically significant differences. For single-qubit systems, this loss function represents the squared geodesic distance on the Bloch sphere, providing a direct geometric interpretation of the optimization landscape.

3.4. *Gradient Estimation via Parameter-Shift Rule*

In variational quantum algorithms (VQA), efficient and hardware-compatible gradient evaluation is crucial for optimizing parameterized quantum circuits. To this end, we employed the parameter shift rule, a native analytic gradient estimation method for QC that yields exact derivatives for a wide range of gates, including all those utilized in our ansatz [35].

For a cost function $\mathcal{L}(\boldsymbol{\theta})$ that depends on a parameterized unitary transformation $U(\boldsymbol{\theta})$, the derivative with respect to an individual parameter θ_k is given by:

$$\frac{\partial \mathcal{L}}{\partial \theta_k} = \frac{1}{2} \left[\mathcal{L}(\boldsymbol{\theta}_+^{(k)}) - \mathcal{L}(\boldsymbol{\theta}_-^{(k)}) \right] \quad (15)$$

where $\boldsymbol{\theta}_{\pm}^{(k)} = (\theta_1, \dots, \theta_k \pm \frac{\pi}{2}, \dots, \theta_M)$ denotes the parameter vector with a shift applied to the k -th component.

This formula holds exactly for gates generated by Pauli operators, such as R_X , R_Y , and R_Z , which are employed in our ansatz. The parameter-shift rule provides theoretical rigour and practical compatibility with quantum technology by removing the requirement for numerical finite differences or sophisticated backpropagation. Since accuracy is of the utmost importance in quantum optimisation, this method is highly advantageous since it produces precise gradients free of approximation errors, in contrast to classical automatic differentiation techniques.

The theoretical foundation of the parameter-shift rule stems from the eigenvalue structure of Pauli generators. For a parameterized gate $U(\theta_k) = e^{-i\theta_k P_k/2}$ where P_k is a Pauli operator with eigenvalues ± 1 , the expectation value of any observable exhibits a sinusoidal dependence on θ_k . This periodic structure allows for the precise calculation of derivatives by means of strategic parameter adjustments, avoiding the numerical mistakes that accumulate in finite-difference approximations [36].

We implemented an improved version of the parameter-shift rule to enhance the computational efficiency and scalability. With a systematic evaluation of the cost function at strategically shifted parameter points $2M$, where M is the total number of variational parameters, our solution computes gradients for all parameters. This batch approach allows for efficient parallelisation and minimises the total number of quantum circuit evaluations required when compared to sequential parameter-by-parameter gradient estimation. This method is feasible for large-scale variational quantum circuits due to the fact that computational complexity scales linearly with the number of parameters.

The gradient computation procedure involves the creation of parameter vectors with systematic shifts applied to the individual components. For each parameter θ_k , we constructed two shifted versions: $\boldsymbol{\theta}_+^{(k)}$ with θ_k increased by $\pi/2$, and $\boldsymbol{\theta}_-^{(k)}$ with θ_k decreased by $\pi/2$. The partial derivative is then computed as the scaled difference between the cost function evaluations at these shifted points. This process is repeated for all parameters to construct the complete gradient vector $\nabla \mathcal{L}(\boldsymbol{\theta})$.

3.5. Quantum-Aware Optimization Strategy

To optimize the variational parameter $\boldsymbol{\theta}$, we introduce a custom quantum-aware optimizer that integrates momentum-based updates with adaptive learning rate scheduling. This method is specifically designed to navigate the complex, nonconvex landscapes characteristic of quantum cost functions, which often exhibit multiple *local minima*, *saddle points*, and regions of vanishing gradients known as *barren plateaus* [37].

The optimization strategy builds upon the classical momentum-based gradient descent while incorporating quantum-specific adaptations [38]. Let $\eta(t)$ denote the learning rate in iteration t , and $\mu \in [0, 1)$ the momentum coefficient. The update rule for the parameter vector $\boldsymbol{\theta}(t)$ follows a two-step process:

$$\mathbf{v}(t+1) = \mu \mathbf{v}(t) - \eta(t) \nabla \mathcal{L}(\boldsymbol{\theta}(t)), \quad \boldsymbol{\theta}(t+1) = \boldsymbol{\theta}(t) + \mathbf{v}(t+1) \quad (16)$$

where $\mathbf{v}(t)$ represents the velocity vector that accumulates momentum from the previous gradient evaluations. The momentum term $\mu \mathbf{v}(t)$ enables the optimizer to maintain direction through regions of inconsistent gradient information, helping to escape shallow local minima and navigate *saddle points* that frequently appear in quantum optimization landscapes.

The adaptive learning rate mechanism is a crucial component of our quantum-aware approach [39]. Conventional fixed-learning-rate strategy frequently falls short in quantum optimisation, as

the optimal step size might fluctuate significantly across various parts of the parameter space. Our adaptive framework modifies the learning rate according to the instantaneous advancements in cost reduction.

$$\eta(t+1) = \begin{cases} 1.05 \cdot \eta(t), & \text{if } \mathcal{L}(t+1) < \mathcal{L}(t), \\ 0.95 \cdot \eta(t), & \text{otherwise.} \end{cases} \quad (17)$$

This adaptive mechanism facilitates swift learning when the cost function consistently declines, signifying that the current optimisation trajectory is effective. Conversely, when costs remain constant or escalate, the learning rate is judiciously diminished to avert overshooting and ensure stability. To prevent oscillatory behaviour that could destabilise the optimisation process, multiplicative factors (1.05 and 0.95) were selected to induce progressive adaptation.

The learning rate adaptation technique tackles many issues inherent to quantum optimisation. A greater learning rate speeds up the convergence in areas with big gradients and good cost function behaviour. Conversely, in proximity to local minima or areas with minimal gradients, the diminished learning rate inhibits the optimiser from executing erratic updates that may impair performance. This dynamic adjustment is especially beneficial when addressing the multiscale characteristics of quantum cost landscapes, where various parameters may necessitate distinct learning rates for optimal convergence.

To ensure numerical stability and prevent divergence, we implemented bounds for adapting of the learning rate. The learning rate was constrained to remain within a specified range $[\eta_{\min}, \eta_{\max}]$, where $\eta_{\min} = 0.1\eta_0$ and $\eta_{\max} = 5\eta_0$ for the initial learning rate η_0 . This constraint prevents the adaptive mechanism from reducing the learning rate to ineffectively small values or from increasing it to potentially unstable levels. The quantum-aware optimizer incorporates additional features to enhance robustness and performance. Early stopping criteria based on fidelity thresholds prevent unnecessary computations when the quality of preparation of the target state is achieved. Specifically, the optimization terminates when the fidelity exceeds 0.9999, which corresponds to a nearly perfect state preparation. This criterion is particularly relevant for QSP tasks, where achieving exact fidelity is the primary objective.

The integration of parameter shift gradients with adaptive momentum-based optimisation provides a robust and scalable framework for developing variational quantum circuits. This methodology is especially efficacious in hardware-aware simulation contexts and NISQ applications, where the integration of precise gradient calculation and adaptive learning ensures both theoretical robustness and practical efficacy [40]. Empirical validation revealed that this quantum-aware optimiser frequently surpassed conventional gradient descent methods and fixed learning rate techniques, attaining quicker convergence and enhanced final solution quality across many QSP problems.

4. Result Analysis

The outcomes of the QML model’s evaluation utilising multiple distance metrics, such as *Trace*, *Fubini-Study*, and *Bures*, are detailed in this section. The outcomes emphasise the compromises that must be made among the quantity of qubits, training duration, accuracy, and the quantity of optimisation factors. We focused on how the model’s performance and optimisation efficiency were affected by raising the qubit count.

Table 1: Performance evaluation of the QNN model with different qubit counts using *Trace distance*.

No. of Qubits	No. of Parameters	Training Time (s)	No. of Iterations	Fidelity
2	18	14.40	73	0.999933
3	27	28.45	72	0.999902
4	36	135.59	200	0.997691
5	45	211.09	200	0.937918

Table 1 illustrates that our custom QNN model performs admirably across various qubit topologies using the *Trace distance*, achieving very high fidelity values generally and revealing the impact of growing circuit complexity. The 2-qubit model’s remarkable performance is a result of its great efficiency and robust learning capability; with only 18 parameters, it achieved the highest fidelity of 0.999933 in only 73 iterations and a relatively fast training period of 14.40 seconds. With 27 parameters and a fidelity of 0.999902, the 3-qubit configuration likewise proves that the model remains very accurate even as the system becomes slightly more complicated, and it again yields outstanding results. When there are 4 qubits, there are 36 parameters instead of 24, but the fidelity stays high at 0.997691. The training time goes up to 135.59 seconds and 200 rounds, but this shows that the model can still learn well in a more intricate optimisation setting. The 5-qubit model, with 45 parameters, reaches a fidelity of 0.937918 after 211.09 seconds and 200 iterations, showing that the model can keep a decent performance even with a higher level of complexity.

Table 2: Performance evaluation of the QNN model with different qubit counts using the *Fubini–Study distance*.

No. of Qubits	No. of Parameters	Training Time (s)	No. of Iterations	Fidelity
2	18	14.46	73	0.999922
3	27	29.03	71	0.999896
4	36	135.14	200	0.999761
5	45	199.54	200	0.921357

Table 2 presents a promising and emphasising insight of our specially designed QNN model’s performance with respect to the *Fubini Study distance*, demonstrating that the model preserves an impressive level of accuracy for the majority of qubit configurations while accurately depicting the effect of heightened circuit complexity. The 2-qubit model achieves the greatest fidelity of 0.999922 with just 18 parameters, 73 iterations, and a brief training period of 14.46s, demonstrating its excellent efficiency and precise learning behaviour. Similarly, the 3-qubit configuration demonstrates exceptional performance; with 27 parameters, the fidelity stays nearly the same at 0.999896, suggesting that the model maintains its excellent learning capabilities despite a modest rise in complexity. In the case of 4-qubits, the number of parameters increases to 36, and the model still attains an extremely precise fidelity of 0.999761. However, it takes the training time 135.14s, and the optimisation process necessitates 200 iterations. This indicates that the model is still quite competent, but

encounters stronger training challenges as the circuit size increases. Considering the system’s significantly increased dimensional complexity, the 5-qubit model still exhibits respectable performance; despite reaching 45 parameters and a fidelity fall to 0.921357 after 199.54s and 200 iterations, the results are still encouraging.

Table 3: Performance evaluation of the QNN model with different qubit counts using the *Bures distance*.

No. of Qubits	No. of Parameters	Training Time (s)	No. of Iterations	Fidelity
2	18	19.29	73	0.999940
3	27	27.76	70	0.999930
4	36	131.49	200	0.999261
5	45	204.02	200	0.942159

Table 3 provides another favourable perspective of our modified QNN model’s performance under the *Bures distance*, demonstrating that the model maintains outstanding fidelity for modest and moderate qubit choices while also emphasising that performance changes as the circuit progressively becomes more complex. With just 18 parameters, the 2-qubit configuration yields the strongest performance, reaching a maximum fidelity of 0.999940 after 73 rounds and a training time of 19.29 seconds. This blatantly exhibits remarkable efficiency and remarkably accurate learning. The 3-qubit model likewise exhibits remarkable performance; this time, with 27 parameters instead of 20, and fidelity kept nearly constant at 0.999930, the model still manages to train in 27.76 seconds with 70 iterations, demonstrating that its remarkable accuracy is preserved despite the slightly more intricate structure. In the 4-qubit case, the number of parameters increases to 36, and the model still achieves a very high fidelity of 0.999261, despite the fact that the training time increases significantly to 131.49 seconds and the optimisation requires 200 iterations, indicating that the larger circuit remains highly capable but becomes more demanding to train. It takes 204.02 seconds and 200 iterations for the model to reach a fidelity of 0.942159 for the 5-qubit configuration using 45 parameters. Although this is lower than the smaller-qubit examples, it still shows a respectable performance considering the significantly larger dimensional complexity.

Table 4: Best-performing quantum-native loss functions based on fidelity score.

Quantum-native loss function	No. of Qubits	No. of Parameters	Fidelity
Bures distance	2	18	0.999940
Bures distance	3	27	0.999930
Fubini–Study distance	4	36	0.999761
Bures distance	5	45	0.942159

Table 4 summarize the exceptional performance of quantum-native loss functions in various qubit setups. The *Bures distance* function performs admirably, attaining 0.999940 with 2 qubits and 18

parameters, 0.999930 with 3 qubits and 27 parameters, and maintaining outstanding results with 5 qubits and 45 parameters, yielding a fidelity of 0.942159. Additionally, the *Fubini Study distance* achieves an outstanding fidelity of 0.999761 with 4 qubits and 36 parameters. The results indicate that our custom QNN framework is both robust and flexible, with *Bures distance* exhibiting consistently strong performance across most qubit configurations, while *Fubini-Study distance* proves effective at 4-qubits, offering insights into the manner in which the optimal loss function may differ based on the number of qubits and the associated parameter complexity.

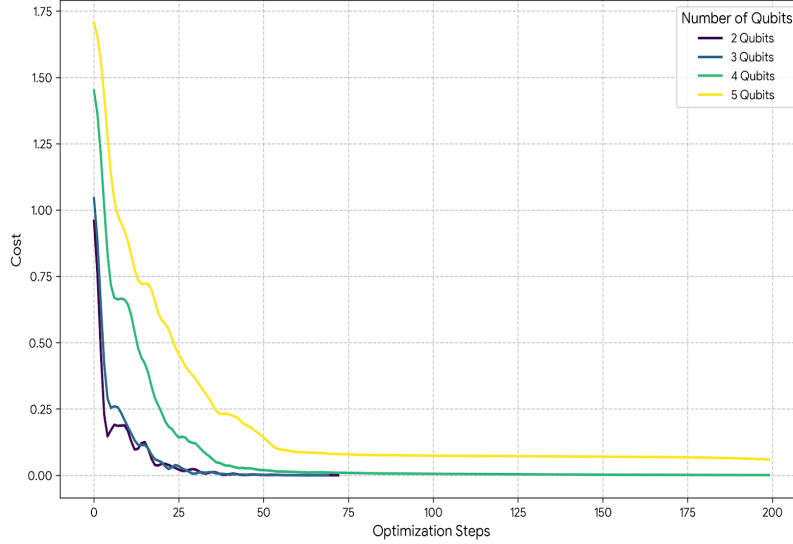


Figure 6: Cost graph for *Bures distance*.

Figure 6 exhibits a promising perspective on the optimisation behaviour of our bespoke QNN model across various qubit configurations, while also distinctly demonstrating how training dynamics evolve with circuit complexity. Our custom QNN framework possesses strong optimisation capabilities, with the 2-qubit and 3-qubit models demonstrating the most efficient convergence, while the 4-qubit and 5-qubit models yield significant insights into the model’s scalability and performance under more demanding training conditions.

Figure 7 shows the particularly designed QNN model’s encouraging optimisation performance across various qubit configurations and precisely and meaningfully shows the effect of growing circuit complexity. Among the models tested, the 2-qubit model shows the most remarkable and rapid learning behaviour; its fidelity increases dramatically from its starting point and reaches almost 1.00 in a short amount of optimisation steps, suggesting good trainability, strong convergence, and efficient parameter usage. The 3-qubit configuration has a parallel and encouraging trend, reaching fidelity near 1.00 as well, albeit with a little more variation in the beginning. This indicates that the model maintains a high level of stability and accuracy even when the complexity is somewhat increased. Even though the 4-qubit model requires more optimisation steps and a bigger parameter space, the fidelity still exhibits a strong increasing tendency and approaches 0.999, indicating that the model can achieve near-perfect performance. Although the 5-qubit configuration starts with the lowest initial fidelity and improves at the slowest rate, it steadily rises during training and settles around

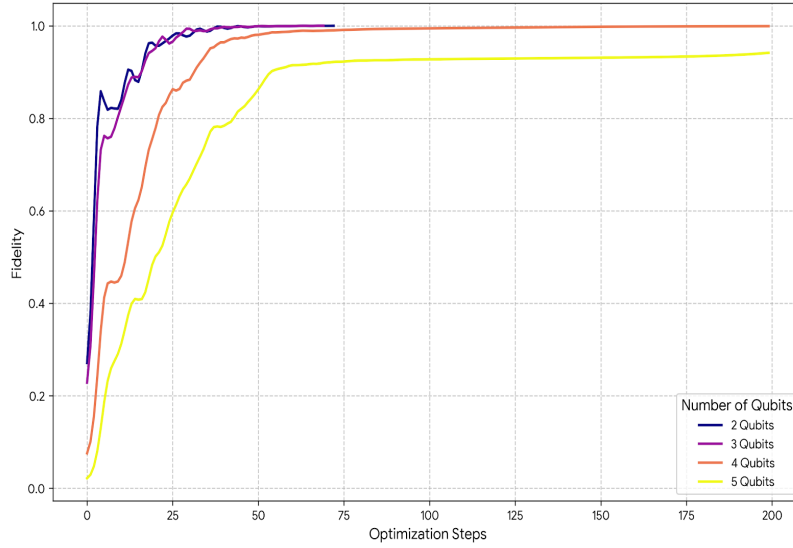


Figure 7: Fidelity graph for *Bures distance*.

0.94, which is still a good result given the system's much higher level of complexity. Overall, the graph demonstrates the QNN framework's resilience, with the 2-qubit and 3-qubit models providing the most effective path to near-perfect fidelity and the 4-qubit and 5-qubit models offering significant insights about the approach's scalability and capacity to sustain strong performance even in more difficult quantum settings.

In this study, the performance of the QNN model was evaluated using several distance metrics, namely the *Trace distance*, *Fubini-Study distance*, and *Bures distance*. Among these, the *Bures distance* demonstrated superior performance in terms of fidelity and optimization efficiency, particularly as the number of qubits increased. Therefore, the subsequent analysis focuses solely on the results obtained using the *Bures distance*. By utilising the *Bures distance*, Figures 8, 9, and 11 illustrate the fidelity behaviour of different qubit systems. Conversely, the fidelity behaviour under the *Fubini-Study distance* is seen in Figure 10. These graphs illustrate the evolution of fidelity across various optimization stages for quantum systems with 2, 3, 4, and 5 qubits.

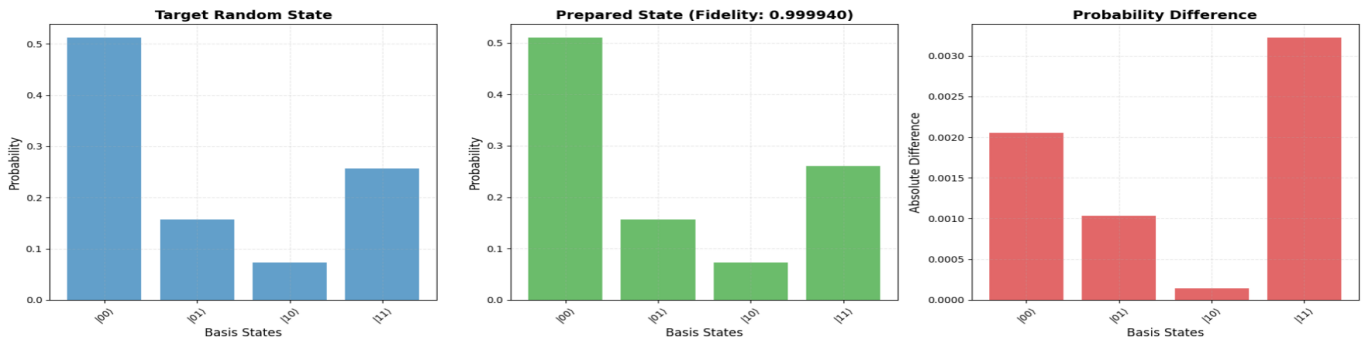


Figure 8: Fidelity graph for *Bures distance* for a 2-qubit system.

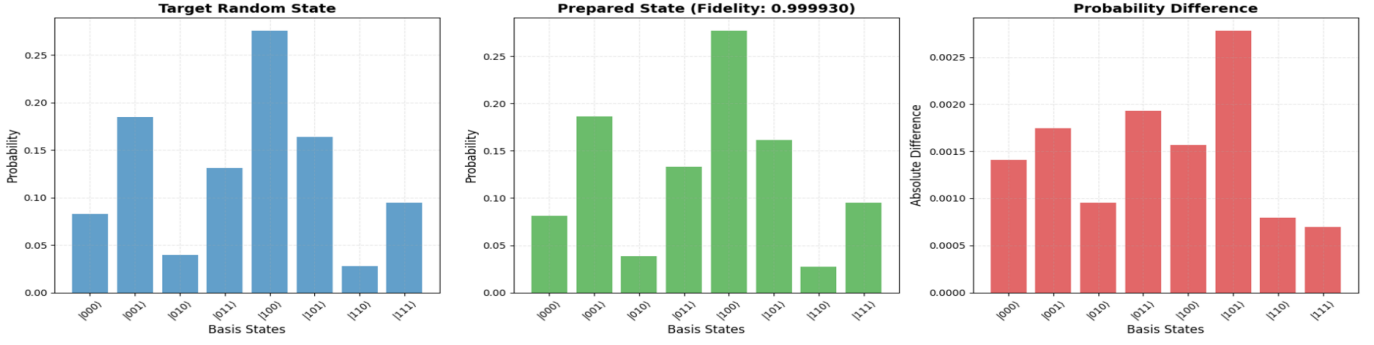


Figure 9: Fidelity graph for *Bures distance* for a 3-qubit system.

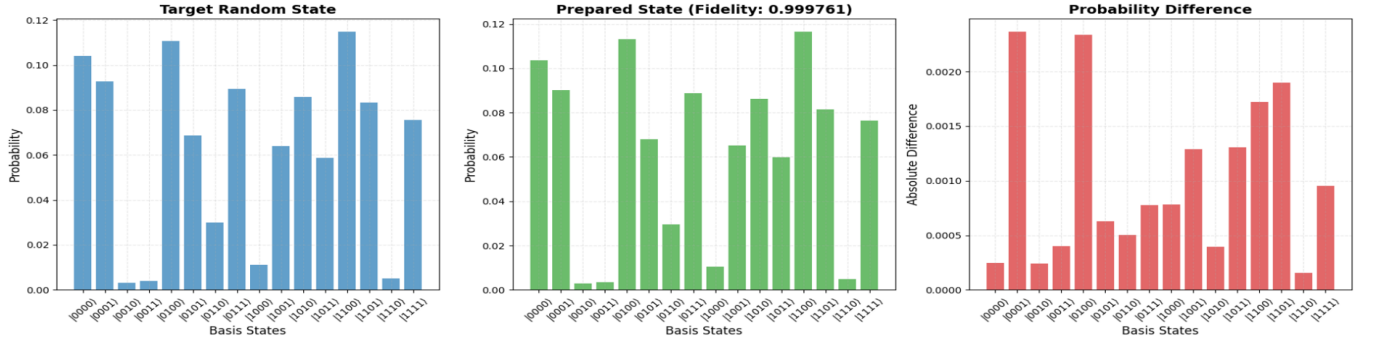


Figure 10: Fidelity graph for *Fubini Study distance* for a 4-qubit system.

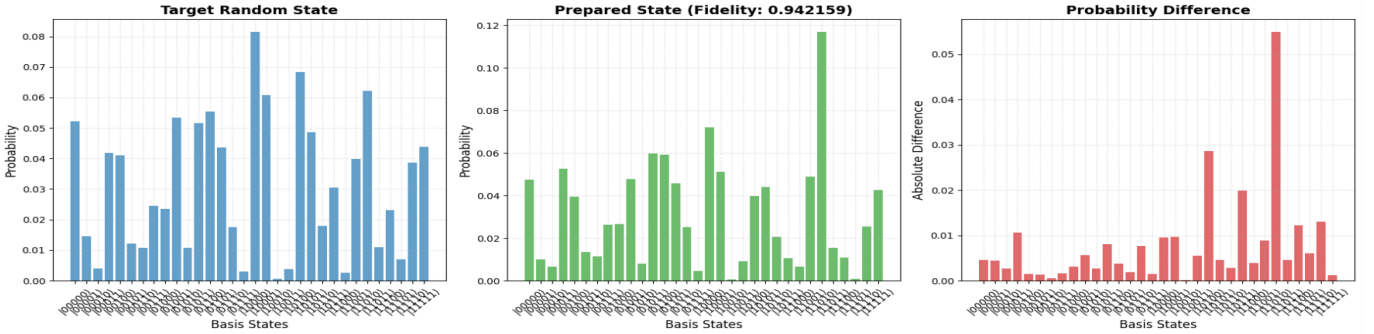


Figure 11: Fidelity graph for *Bures distance* for a 5-qubit system.

The probability distributions for randomly generated quantum states with 2 two 5-qubits are shown in Figure 8-11. Figure 8 indicates that the 2-qubit state has an extremely high fidelity of 0.999940, which means that there are almost no variations between the target and the prepared states and that they are almost identical. In Figure 9, the 3-qubit state, where the prepared state is similar to the target state, has a fidelity of 0.999930, and no significant differences are visible. With relatively few changes in some of the basis states, the 4-qubit state in Figure 10 attains a fidelity of, indicating a very precise preparation. Figure 11 shows a 5-qubit state with a fidelity

of 0.942159. Greater differences in particular basis states indicate areas in which state preparation should be strengthened. In general, the data show that state preparation is of high quality, with a few exceptions in higher-qubit states.

5. Conclusion

This paper introduces a QSP framework that leverages a QNN in a VQA architecture to prepare high-fidelity quantum states for NISQ devices. Our novel variational ansatz with linear parameter scaling balances circuit expressivity with trainability, making it scalable for large quantum systems and high-performance on NISQ technology. To improve the convergence speed of the optimization processes and overall robustness, quantum native loss functions such as Trace distance, Bures distance, and Fubini-Study distance are used. These functions optimize the preparation of quantum states by considering the intrinsic geometry of quantum state spaces. In addition, the enhanced parameter shift rule for precise gradient computation and the quantum-aware adaptive optimizer with momentum-based updates reduce the processing overhead in QSP problems' complicated and nonconvex optimization landscape tasks. Among the three distance metrics evaluated, the Bures distance had the best fidelity and optimization efficiency for 2-qubits, 3-qubits, 5-qubits, and the Fubini-Study distance had the best fidelity and optimization efficiency for 4-qubits. Experimental results show that training time grows with qubit count, from 19.29s for 2-qubits to 204.02s for 5-qubits. The fidelity of 0.999940 for 2-qubits, 0.999930 for 3-qubits, and 0.999261 for 4-qubits shows that smaller qubit systems still have good quantum state preparation. The fidelity lessens with increasing qubits, reaching 0.942159 for 5-qubits. Because of its superior performance, noise control capabilities, and scalability to larger systems, the Bures distance has been proposed as the best cost function for our quantum state preparation technique, while the Fubini-Study distance also proves to be a competitive choice, further supporting the effectiveness of these quantum-native loss functions in optimizing quantum machine learning models.

The framework has a lot of potential; however, there are a number of areas that need to be improved. Firstly, future research will expand the experiment to incorporate real quantum hardware. In addition, to reduce quantum noise and increase short-term device performance, we propose the use of quantum error correction approaches, including surface coding and topological error mitigation. Another approach involves using quantum feature maps to enhance expressivity, particularly when generating complex quantum states in fields such as quantum chemistry and drug development. Additionally, our method can be combined with quantum optimization techniques such as QAOA to enhance state preparation.

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